

Dutch Road-Traffic CO₂ Emissions: A Reproducible Sustainability Regression Project

Project Overview & Problem Statement

This project focuses on a sustainability centered regression problem: predicting annual carbon dioxide (CO₂) emissions from road traffic in the Netherlands. The goal is to understand how road-traffic activity and vehicle fleet characteristics influence national transport-related emissions over time, using official Dutch government data from CBS StatLine.

The selected target variable is annual CO₂ emissions from road traffic on Dutch territory, measured in million kilograms of CO₂. This outcome is important because transport remains a significant contributor to greenhouse gas emissions, and road traffic is one of the most policy-relevant sources for climate mitigation through cleaner vehicles, improved efficiency, and transport planning.

This prediction problem matters because a reliable regression model can help identify which factors are most strongly associated with emissions changes. In a sustainability context, those insights can support more informed decisions about fleet efficiency improvements, transport decarbonisation strategies, and long-term climate policy evaluation. Reproducibility is especially important here because sustainability models may inform public decisions and therefore must be transparent, auditable, and easy to rerun.

Problem Statement:

How do changes in road traffic activity and vehicle fleet characteristics influence annual CO₂ emissions from road traffic in the Netherlands?

(To Predict annual CO₂ emissions from road traffic in the Netherlands based on road-traffic activity and vehicle fleet characteristics.)

Hypothesis

This project is based on the hypothesis that annual CO₂ emissions from Dutch road traffic are positively associated with road-traffic activity and negatively associated with improvements in average fleet efficiency. In other words, when more vehicle kilometres are travelled, CO₂ emissions are expected to increase, while lower fleet emission factors in grams per kilometre are expected to reduce total emissions.

H1: Higher road-traffic activity is associated with higher annual CO₂ emissions from road traffic in the Netherlands.

H2: Lower average fleet emission factors for CO₂ are associated with lower annual CO₂ emissions, holding other factors constant.

H3: Changes in the composition of the vehicle fleet, such as the share of heavy commercial vehicles relative to passenger cars, are associated with measurable differences in annual road-traffic CO₂ emissions. The CBS table shows large variation in fleet emission factors across categories, with passenger cars at 143 grams/km in 2024 and heavy commercial motor vehicles at 882 grams/km in 2024, which supports the idea that vehicle composition may matter for total emissions.

Dataset Selection

The main dataset selected for this project is the CBS StatLine table **“Emissions to air on Dutch territory; road traffic”** (table identifier: 85347ENG), published by Statistics Netherlands (CBS). The dataset is available on the CBS website and reports annual emissions from road traffic on Dutch territory, including emissions from foreign vehicles, as well as average fleet emission factors expressed in grams per driven kilometre.

Source **organization:** Statistics Netherlands (CBS)
Dataset title: Emissions to air on Dutch territory; road traffic
URL: <https://www.cbs.nl/en-gb/figures/detail/85347ENG>

This dataset is well aligned with the project because it includes both the main environmental outcome of interest and explanatory variables related to vehicle efficiency and vehicle type. The table explanation states that emissions are calculated using activity data, such as vehicle kilometres, multiplied by emission factors, which makes the dataset conceptually appropriate for a regression-based emissions analysis.

The selected target variable is:

- Annual CO₂ emissions from road traffic on Dutch territory (million kg CO₂) for Total motor vehicles.

This is a continuous quantitative variable, which makes it appropriate for regression modeling. It directly reflects a sustainability outcome of interest because CO₂ is a greenhouse gas formed through fuel combustion and is central to climate change mitigation.

Basic Dataset Information

The selected CBS StatLine table contains annual observations and has a reference period from 1990 to 2024. According to the table information, actual emissions are available from 1990 onward, while fleet emission factors are available from 2004 onward. The dataset is updated yearly, and the latest provisional figures for 2024 were added on 5 February 2026.

The geographic scope is the Netherlands, specifically Dutch territory, and the table includes emissions generated by road traffic, including foreign vehicles using Dutch roads. The dataset includes multiple vehicle categories and multiple pollutants, but this project will focus on the CO₂-related variables.

The file information shown on the dataset page indicates that the table contains 245 rows and 32 columns, which makes it manageable for inspection, cleaning, and regression modeling in a notebook environment.

Ethical, legal, and licensing considerations

This dataset is appropriate for a sustainability capstone from an ethical and legal perspective because it contains aggregated environmental statistics rather than personal or sensitive individual-level data. That means the project does not raise direct privacy concerns such as identification of persons, households, or protected groups.

From a legal and reuse perspective, the CBS page links to an open licence statement and provides public access through StatLine and APIs. This makes the dataset suitable for educational and analytical use, provided the source is properly cited and documented.

From an ethical perspective, it is still important to communicate the limits of interpretation. A regression model based on national annual data can reveal patterns and associations, but it cannot by itself prove policy causality. The project should therefore avoid overstating conclusions and should clearly report assumptions, data limitations, and reproducibility steps.

Dataset Understanding and Early Troubleshooting

A major part of the project journey was understanding the raw dataset correctly. The first attempt to load the CSV produced a dataframe with shape (245, 1), which showed that the file had been parsed incorrectly as a single column. This happened because the CSV was semicolon-separated, so the file had to be reloaded using `pd.read_csv(file_path, sep=";")`, after which the dataframe loaded correctly with shape (245, 33).

The notebook also showed that some practical file-management issues had to be solved before analysis could begin. The raw file was initially not found because the filename ended in `.csv.csv`, which caused a file path mismatch. After fixing the filename and reading the file with the correct separator, the column structure became visible and the project could move on to data preparation.

Data Preparation

The first stage of cleaning focused on structural preparation rather than advanced preprocessing. Exact duplicates were checked, and the notebook reported Exact duplicates: 0, so no duplicate rows needed to be removed. The Periods column was then

converted into a numeric year feature by taking the first four characters of each period string and converting them to integers.

An early assumption was that MeansOfTransport would contain human-readable labels such as “Total motor vehicles,” but the notebook revealed that it actually contained categorical codes such as A018928, A018943, and similar identifiers. Because filtering on a text label returned zero rows, that approach was abandoned and all available transport-category rows were kept in the working dataframe. This was an important lesson in checking data semantics before applying filters.

A smaller modelling dataframe was then created with four columns: year, MeansOfTransport, co2_emissions_mln_kg, and co2_fleet_gram_per_km. The target variable came from CarbonDioxideCO2_1, and the fleet-intensity feature came from CarbonDioxideCO2_16. These fields were renamed to make the notebook more readable and to support later modelling steps more clearly.

The next important preparation step involved dealing with non-numeric placeholders. The notebook showed that co2_fleet_gram_per_km contained values such as "." rather than valid numbers, while the emissions column was already numeric. Both columns were converted with `pd.to_numeric(..., errors="coerce")`, which turned invalid strings into missing values, and incomplete rows were then dropped. After this step, the cleaned modelling dataframe had shape (147, 4), which became the final dataset used for regression.

Model Training

After cleaning, the dataset was split into training and test sets for regression. The final model used co2_fleet_gram_per_km and year as features and co2_emissions_mln_kg as the target variable. After dropping rows with missing feature values, the split produced 117 training rows and 30 test rows.

A linear regression model from scikit-learn was then trained on the training set. Earlier in the workflow, the environment did not yet contain scikit-learn, so the package had to be installed before the model could be run successfully. Once the package issue and the non-numeric data problem were resolved, the regression model executed correctly and produced coefficients and performance metrics.

Evaluation Results

The trained model returned an intercept of 256885.83 and two coefficients: -5.78 for co2_fleet_gram_per_km and -122.74 for year. In practical terms, this means the fitted model estimated lower emissions both for later years and for higher fleet emission factors, which is partly counterintuitive and signals that the model is not capturing the underlying relationships very well. The negative year coefficient is plausible because

emissions may decrease over time, but the negative fleet-factor coefficient contradicts the expected environmental logic.

The performance metrics confirm that the model had very weak predictive power. The training RMSE was 9524.59 and the test RMSE was 10256.99, while the training R^2 was 0.039 and the test R^2 was 0.002. These values indicate that the model explained almost none of the variation in annual CO₂ emissions and that it performed only marginally better than a naive baseline.

Key Findings

One important finding is that the technical workflow itself worked by the end of the notebook. The data was successfully loaded, cleaned, transformed into a modelling dataframe, split into train and test sets, and used to fit a regression model. Even though the predictive results were weak, the modelling pipeline became functional and reproducible.

A second finding is that the selected predictors were not sufficient to explain the target well. Using only year and average fleet CO₂ grams per kilometre did not capture enough of the real-world drivers of total road-traffic emissions. This likely happened because total emissions also depend heavily on traffic activity, travel demand, and the composition of different transport modes, which were not properly represented in the final simple model.

A third finding is that real-world public datasets require careful inspection before modelling. The project encountered multiple realistic issues: file naming errors, a separator mismatch, coded category labels instead of expected text labels, and non-numeric placeholder values. Solving those problems was a significant part of the regression journey and demonstrates practical data understanding, not just model building.

What Worked

The strongest part of the project was the data workflow. The notebook correctly identified and solved several loading and preparation issues, including wrong file naming, incorrect CSV parsing, and hidden missing values represented by ".". The final cleaned dataframe and train-test split show that the project reached a valid modelling stage rather than stopping at raw data inspection.

Another aspect that worked well was the use of clear variable renaming and project structure. Renaming columns such as CarbonDioxideCO2_1 to co2_emissions_mln_kg improved readability, and saving a cleaned file into data-processed supported reproducibility and later reuse. These are good habits for machine learning projects and make the analysis easier to explain in a presentation.

What Did Not Work

The main thing that did not work well was the predictive performance of the model itself. The linear regression with only two features did not explain the target meaningfully, and the near-zero R^2 values show that the model was under-specified. This means the problem was not simply overfitting; instead, the model lacked enough relevant explanatory variables.

Another issue was that the initial interpretation of MeansOfTransport was incorrect. The project first assumed that this field contained readable category names, but it actually contained codes, so filtering for “Total motor vehicles” removed all rows. Although this was later corrected, it delayed the analysis and highlighted the importance of validating assumptions against the raw data.

Why the Results Looked Weak

The weak results make sense when considering the structure of the problem. Total annual CO₂ emissions are influenced not only by vehicle efficiency, but also by how much traffic occurs, what types of vehicles are on the road, and how transport demand changes over time. A model using only year and one fleet-efficiency variable is therefore too simple to represent the system realistically.

There may also be confounding between the two features that were used. Over time, fleet emission factors may decline while total emissions are also affected by independent macro changes such as economic activity, mobility trends, or policy interventions. In that situation, the model can assign coefficients that are mathematically fitted but difficult to interpret substantively, which is exactly what seems to have happened here.

Best Model Performance Metrics

The best reported model in the notebook was the linear regression using `co2_fleet_gram_per_km` and `year` as predictors. Its performance metrics were: Train RMSE = 9524.59, Test RMSE = 10256.99, Train R^2 = 0.039, and Test R^2 = 0.002. These are the numbers to bring to the presentation when asked for model performance.

One Surprising Insight

One surprising insight was that the fleet emission factor coefficient was negative. Intuitively, higher grams of CO₂ per kilometre should be associated with higher total emissions, but the model estimated the opposite relationship. This was a useful warning sign that the simple model was not capturing the underlying transport-emissions system correctly and that coefficient signs should always be interpreted critically rather than accepted automatically.

One Challenge and How It Was Solved

One major challenge was that the dataset appeared broken at first. The CSV initially loaded as only one column because the wrong delimiter was used, and later the model failed because placeholder strings such as "." were stored inside a numeric feature column. These issues were solved by reloading the file with `sep=";"`, converting relevant columns with `pd.to_numeric(..., errors="coerce")`, and dropping rows with missing values after coercion.